

A Machine-Learning Ensemble for Probabilistic Influenza Forecasting Across Nigeria: Using LightGBM and Random Forest Algorithms

Individual Category: Dexwel

Abstract

Background: Seasonal influenza places a recurring burden on Nigeria's health system, made harder by uneven healthcare access and a climate pattern that does not match the single winter peak seen in temperate countries. Early, reliable forecasts, accompanied by realistic uncertainty ranges rather than a single point estimate, can help hospitals and health officials prepare before cases rise.

Objective: This study develops and evaluates a machine-learning ensemble for forecasting weekly influenza cases, hospitalisations and deaths across Nigeria's 36 states and the Federal Capital Territory, and tests, rather than assumes, which modelling approach is best suited to this task.

Methods: One pooled model per outcome was built for three seasons of weekly surveillance data, predicting up to eight weeks ahead from lagged and rolling case history, a short-term growth ratio, calendar features, and each state's population, geopolitical zone and climate class. Probabilistic forecasts were produced by combining LightGBM quantile-regression models with a Random Forest, averaged with equal weight. Model suitability was evaluated empirically: a seasonal-naive baseline, a Random Forest, a scikit-learn Gradient Boosting model, LightGBM alone, and the proposed ensemble were trained on identical data and scored under an identical rolling-origin protocol.

Results: All machine-learning models reduced mean absolute error (MAE) to approximately one-third of the naive baseline's. On the largest evaluation fold, the ensemble achieved the lowest MAE, Weighted Interval Score (WIS*) and Continuous Ranked Probability Score (CRPS) for cases, hospitalisations and deaths (cases: MAE 81.0 versus 82.6 for LightGBM alone; WIS* 174.6 versus 186.0), and across both rolling-origin test seasons it reduced WIS* by 9-11% relative to LightGBM alone. The ensemble did not outperform every alternative on every criterion: LightGBM alone achieved a lower RMSE for cases, and scikit-learn's Gradient Boosting achieved better-calibrated interval coverage. Mean 90% interval coverage for cases across both test seasons was 50.4%, rising from 14.0% to 86.7% once a second season of training data became available.

Conclusion: The equal-weight LightGBM-Random Forest ensemble is proposed as the deployed model on the basis of its accuracy and probabilistic performance, not because it dominates every evaluation metric. Conformal calibration and periodic retraining as new seasons accumulate are identified as priorities for future refinement. The complete pipeline, comparison experiments and a deployable forecasting service are reproducible from the supplied surveillance data.

Keywords: *influenza forecasting; machine learning; gradient boosting; random forest; ensemble learning; probabilistic forecasting; Nigeria; public health surveillance*

Source code (GitHub): <https://github.com/maepimschallenge/MAEPiMS-2.0-2026-/tree/main/Dexwel>

Live forecasting demonstration : <https://flu-forcaster.streamlit.app/>

1. Introduction

Seasonal influenza represents a major, under-recognized cause of acute respiratory illness and population morbidity across Sub-Saharan Africa. In tropical and subtropical regions, the epidemiological dynamics of influenza deviate significantly from the well-defined single-winter peaks characteristic of temperate northern latitudes. In West Africa, disease transmission exhibits complex, heterogeneous temporal patterns driven by subtle climatic variations, shifting precipitation regimes, and complex socioeconomic dynamics. In Nigeria—Africa’s most populous nation—seasonal influenza places a persistent, recurring burden on an already strained healthcare infrastructure. The nation’s unique geographical breadth, spanning tropical monsoon climates in the southern coastal regions to arid Sahelian zones in the far north, yields diverse regional micro-climates that profoundly influence virus survival, transmission efficiency, and population exposure dynamics (Akano et al., 2024; Lagare et al., 2025).

The public health challenges posed by seasonal influenza in Nigeria are substantially amplified by widespread structural healthcare disparities. Access to primary healthcare, diagnostic infrastructure, and essential medical countermeasures remains deeply uneven across administrative boundaries. Rural communities and underserved urban informal settlements frequently bear a disproportionate burden of severe disease outcomes, exacerbated by delayed clinical presentation, limited capacity for laboratory confirmation, and supply chain vulnerabilities for critical medical commodities such as antivirals and medical oxygen support. Furthermore, high population mobility across geopolitical zones, driven by internal trade, urbanization, and agricultural migrations, facilitates rapid spatial dissemination of respiratory pathogens. When seasonal surges coincide with peaks of other endemic infectious diseases—such as malaria, Lassa fever, or severe acute respiratory infections of alternative etiologies—the resulting operational pressure on emergency departments and intensive care units can severely compromise overall health system resilience.

Surveillance efforts across West Africa have increasingly highlighted the role of environmental factors in shaping regional epidemic timing. Multi-year epidemiological studies in neighbouring countries, such as Côte d'Ivoire and Niger, demonstrate that influenza activity is closely aligned with local precipitation, relative humidity, and seasonal harmattan dust storms (N'gattia et al., 2016; Lagare et al., 2025). In coastal West Africa, peak transmission frequently corresponds with heavy rainfall during the primary rainy season, whereas interior Sahelian areas often experience secondary viral revivals during cooler, drier months (N'gattia et al., 2016; Lagare et al., 2025). Secondary analyses of national sentinel surveillance in Nigeria confirm multi-wave annual activity patterns that vary substantially across ecological and geopolitical boundaries (Akano et al., 2024). However, routine passive surveillance alone offers limited forward-looking utility for public health planning, as reported case counts inherently reflect past transmission events rather than future trajectory.

To bridge the gap between retrospective surveillance and proactive outbreak response, advanced public health decision-support systems rely heavily on prospective epidemiological forecasting. Timely, reliable predictions of upcoming epidemic trajectory enable health ministries and emergency operations centres to optimize resource allocation weeks in advance. Specifically, accurate forward projections support the strategic pre-positioning of medical supplies, the dynamic adjustment of clinical staffing schedules, targeted public health communication campaigns, and the prioritized deployment of outreach intervention units to high-risk administrative zones. Moreover, in modern public health practice, point

forecasts alone are insufficient; decision-makers require honest, calibrated representations of predictive uncertainty to weigh the operational costs of early intervention against the potential risks of under-preparedness (Reich et al., 2019; Bracher et al., 2021; Ray et al., 2023).

Despite the recognized value of predictive surveillance, rigorous multi-horizon epidemiological forecasting tailored specifically to Nigeria's administrative and geographic layout remains sparsely developed. Most existing literature focuses primarily on retrospective descriptive epidemiology, characterising past seasonal trends or global phylogenetic lineages rather than providing actionable, forward-looking operational risk assessments (Akano et al., 2024). Furthermore, multi-region forecasting across developing nations faces distinct challenges, including spatial reporting inconsistencies, non-stationary clinical healthcare seeking behaviours, and pronounced regional demographic shifts. There is a pressing necessity for formal predictive frameworks that can systematically capture complex temporal dependencies, seasonal momentum, and regional vulnerabilities simultaneously across diverse administrative domains.

Addressing these critical gaps is vital for establishing robust public health early warning systems capable of operating effectively within resource-constrained healthcare environments. The primary objective of this study is to establish and rigorously evaluate a spatial-temporal probabilistic forecasting framework for weekly influenza cases, hospitalisations, and mortality across administrative divisions in Nigeria. By generating detailed multi-horizon projections accompanied by explicit probabilistic uncertainty ranges, this study aims to provide public health administrators and policymakers with a reliable, evidence-based decision-support tool for pre-emptive epidemic planning and resource optimization.

2. Assumptions

The linelist were used to build, evaluate and compare the candidate models; no external health, climate or mobility data were introduced. The following assumptions underlie the modelling approach:

- i. Weekly reported cases, hospitalisations and deaths are treated as noisy, discretely reported counts. Recent history, specifically the preceding four weeks, expressed through lagged values and rolling statistics, is assumed to carry most of the short-term signal relevant to an eight-week forecast horizon.
- ii. Cases, hospitalisations and deaths are modelled on a log1p scale, standard practice for heavily right-skewed epidemic count data, since a small number of very large outbreak weeks would otherwise dominate a model trained on the untransformed scale.
- iii. The forecast horizon (1 to 8 weeks ahead) is included as an explicit numeric feature within a single pooled model rather than through eight separately trained horizon-specific models, on the assumption that one model can learn how the forecast should vary with horizon at least as effectively, with substantially reduced implementation complexity.
- iv. A state's population, geopolitical zone, climate class and healthcare-access score are assumed to be informative, largely time-invariant covariates. The model is not supplied with geographic coordinates or a mobility matrix, as neither is available in the source data.
- v. The calendar position of both the forecast origin and the forecast target week is assumed to be informative and must be computed identically wherever the model is trained or deployed; this study specifically identified an instance in which failing to enforce this identically corrupted

multi-week-ahead forecasts without altering apparent one-step accuracy (see Model Architecture).

- vi. The two constituent model families, boosting and bagging, are combined with equal, untuned weights. With only three seasons of data available, there is insufficient history to estimate trained ensemble weights reliably, and untrained equal-weight averages are recognised as a robust default when the relative performance of the constituent models is unstable (Ray et al., 2023).
- vii. Because independently trained quantile models are not guaranteed to yield a monotonically increasing sequence of quantiles for a given input, predicted quantiles are sorted into monotonic order as a standard post-processing step.
- viii. National- and zone-level forecast ranges are constructed by summing state-level forecasts, both the mean and each quantile independently. This procedure assumes that states move in close to lockstep.

3. Forecasting Methodology

The analysis proceeds in eight stages, each implemented as a discrete, independently inspectable step in the accompanying source code. First, the five supplied data files are loaded and validated; Second, a feature table is built in which every row corresponds to one state, one forecast origin week, and one forecast horizon (1 to 8 weeks ahead), with features drawn exclusively from information available at or before the origin week. Third, for each outcome (cases, hospitalisations, deaths) a LightGBM model is trained with squared-error loss for the point forecast, together with nine further LightGBM models trained with pinball loss at the quantile levels 0.025 through 0.975. Fourth, a Random Forest is trained per outcome on the same engineered features, one-hot encoded for the categorical zone and climate variables. Fifth, the two components' predictions are combined by simple averaging, in original (back-transformed) units, with the resulting quantiles sorted into monotonic order to form the deployed ensemble forecast. Sixth, state-level forecasts are aggregated to North, South and national geographies by summation. Seventh, a Monte Carlo simulation drawn from the ensemble's quantile forecasts yields the optional probabilistic targets requested by the Challenge: the probability that each horizon week holds the season peak, and the probability that transmission is increasing rather than decreasing from one week to the next. Eighth, the entire pipeline is evaluated by strictly chronological rolling-origin backtesting: for each held-out season, every model is refit using only the seasons preceding it, and forecasts at each origin week use only information that would genuinely have been available at that point in time.

This staged structure means every reported figure and table can be traced to a single, named step in the source code, and the same procedure that produces the backtested evaluation also produces the live eight-week forecast reported in the Results section, with no separate, untested code path for either.

4. Model Architecture

4.1 Features

For every state and every historical week, a feature vector was constructed exclusively from information available prior to the forecast target: four weeks of lagged values, a rolling four-week mean and standard deviation, and a short-term growth ratio, each computed separately for cases, hospitalisations and deaths; the state's log-transformed population, geopolitical zone and climate class; a sine-cosine pair

encoding the calendar position of the forecast origin; a second sine-cosine pair encoding the calendar position of the forecast target week; and the forecast horizon itself, ranging from one to eight weeks. All count-based features and targets were transformed using $\log_{1p}$ prior to training and transformed back via the inverse function thereafter:

$$y' = \log(1 + y), \quad \hat{y} = \exp(\hat{y}') - 1$$

A methodological lesson in feature-pipeline consistency

An early version of this pipeline computed the two calendar sine-cosine features from the forecast origin week during training, but from the forecast target week when generating a live forecast. As these two values coincide only when the horizon equals one week, the resulting inconsistency was undetectable in ordinary one-step accuracy checks yet caused multi-week-ahead forecasts to diverge to implausible values, because the model had never been trained to interpret a target-week calendar signal that varies independently of the horizon. The remedy did not involve any change to the model itself: it consisted of implementing the feature-construction logic exactly once, in a module shared by both the training script and the deployed forecasting service, so that the two could not subsequently diverge. The same discipline was applied to the categorical variables: geopolitical zone and climate class are always encoded against a fixed category list, ensuring that a single-row forecast request is encoded identically to the full training table.

4.1 LightGBM component

LightGBM (Ke et al., 2017) is a gradient-boosted decision-tree ensemble in which an initial prediction is iteratively refined by successive shallow decision trees, each fitted to the residual errors of its predecessors. One model was trained per outcome using ordinary squared-error loss to obtain the point (mean) forecast, and nine further models per outcome were trained using the pinball (quantile) loss (Koenker & Bassett, 1978),

$$L_{\tau}(y, \hat{y}) = \max(\tau(y - \hat{y}), (\tau - 1)(y - \hat{y}))$$

at the quantile levels 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95 and 0.975, yielding a full predictive distribution rather than a single point estimate. Hyperparameters (31 leaves per tree, a learning rate of 0.05, 300 boosting rounds, 80% row and column subsampling per tree, and a modest L2 penalty) were fixed at conventional, literature-typical values rather than optimised through an exhaustive search, since an aggressive search conducted on only three seasons of data risks overfitting to the particular seasons available.

4.2 Random Forest component and ensemble construction

The Random Forest component (Breiman, 2001; scikit-learn implementation, Pedregosa et al., 2011) comprises 100 trees, a minimum of five observations per leaf, and a per-split feature subsample equal to the square root of the total feature count, trained on the same $\log_{1p}$ -transformed targets with geopolitical zone and climate class one-hot encoded. One forest was trained per outcome; its point forecast is the mean prediction across constituent trees, and its quantiles are derived from the empirical spread of the individual trees' predictions, a simplified analogue of a quantile regression forest (Meinshausen, 2006).

The deployed forecast is the equal-weight average of the LightGBM and Random Forest predictions, computed in original units after each component's predictions are back-transformed, with the resulting quantiles subsequently sorted into monotonic order. No ensemble weights are tuned, which keeps the combination simple, computationally inexpensive, and resistant to overfitting.

4.3 Hospitalisation and death channels

Hospitalisations and deaths are forecast on the same per-100,000-population scale as cases, using the same feature set and the same two-component ensemble structure, rather than being derived as a fixed ratio of the case forecast; this allows each outcome's own lagged history, growth rate and seasonal signal to inform its forecast directly.

4.4 Evaluation metrics and protocol

All models were evaluated under a strictly chronological protocol: each was refitted using only the seasons preceding the designated test season, ensuring that no future information could leak into training (Bergmeir & Benitez, 2012). In addition to mean absolute error (MAE) and root-mean-squared error (RMSE) for the point forecast, two proper scoring rules were used to evaluate the full predictive distribution. The Weighted Interval Score (Bracher, Ray, Gneiting, & Reich, 2021) scores a set of central prediction intervals against the observed value y ; for a single interval at level α with bounds l and u ,

$$IS_{\alpha}(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)1\{y < l\} + \frac{2}{\alpha}(y - u)1\{y > u\}$$

and the variant reported here, WIS^* , averages this quantity across the 90%, 80% and 50% central intervals together with the absolute error of the median forecast m :

$$WIS^* = \frac{1}{3.5} \left[\frac{1}{2}|y - m| + \frac{1}{2} \sum_{\alpha \in \{0.1, 0.2, 0.5\}} IS_{\alpha}(l_{\alpha}, u_{\alpha}; y) \right]$$

A quantile-based approximation to the Continuous Ranked Probability Score (Gneiting & Raftery, 2007) was also computed from the same quantile grid:

$$CRPS \approx \frac{2}{K} \sum_{k=1}^K (\tau_k - 1\{y < q_{\tau_k}\})(y - q_{\tau_k})$$

Both scoring rules reward forecasts that are simultaneously accurate and appropriately narrow, such that a model cannot improve its score merely by widening its intervals until they invariably contain the observed value.

4.5 Computational environment and software

All analyses were conducted in Python using pandas (McKinney, 2010), NumPy (Harris et al., 2020), scikit-learn (Pedregosa et al., 2011), LightGBM (Ke et al., 2017) and Matplotlib (Hunter, 2007). The

complete source code, the trained model, and a deployable forecasting service accompany this report (see GitHub above, and the streamlit link).

5. Model Justification

Three broad modelling strategies were available for this task: a mechanistic model of person-to-person transmission, a classical statistical time-series model, and a machine-learning model fitted directly to the historical pattern in the data. A mechanistic model was not adopted because it requires epidemiological parameters, such as a generation interval and contact rates, that cannot be recovered from weekly aggregate totals alone, and fitting such parameters purely to match three seasons of aggregate incidence risks non-identifiability and false precision. A classical statistical model, for example a seasonal ARIMA fitted separately for each state, was not adopted as the primary approach because it does not naturally pool information across the 37 reporting units and cannot readily incorporate static covariates such as geopolitical zone or healthcare access. Tree-ensemble machine-learning models address both requirements directly: a single pooled model learns jointly across all states, weeks and forecast horizons, while combining numeric time-series features with categorical state metadata in one coherent framework.

Within the class of machine-learning models, tree ensembles were preferred to deep neural networks because the available training data, three seasons, 37 states and 52 weeks, is limited for deep-learning purposes, whereas tree ensembles are known to perform strongly on small-to-medium tabular datasets of this kind (Makridakis, Spiliotis, & Assimakopoulos, 2022). Two tree-ensemble families were subsequently combined because they err in different ways: boosting constructs trees sequentially to correct the residual errors of preceding trees, whereas a Random Forest averages a large number of independent, deliberately decorrelated trees. Averaging forecasts produced by models whose errors are only partially correlated is known to reduce overall error, and this result is among the most consistently replicated findings in the forecasting literature (Clemen, 1989; Timmermann, 2006). Rather than assume this benefit transfers to the present setting, the study quantified it directly through the model-comparison experiment reported below, in which a seasonal-naïve baseline, a Random Forest, scikit-learn's Gradient Boosting, LightGBM and the proposed ensemble were trained and scored identically.

The ensemble was ultimately selected for deployment because it achieved the lowest MAE, WIS* and CRPS of all five candidates for every outcome on the largest test fold, at a modest additional computational cost of approximately seven seconds of training time relative to LightGBM alone. This is a more measured justification than an assumption that gradient boosting, or any single model family, is inherently best for epidemic forecasting: the choice is grounded in a same-data comparison reported in full in the Results section, including the respects in which the ensemble did not outperform every alternative.

6. Results

6.1 Descriptive overview of the supplied seasons

Across the three seasons represented in the data, the mean national case rate rose from a moderate level in 2023/2024 to a high level in 2024/2025, before declining to a moderate-to-high level in 2025/2026 (Table 1), consistent with the severity labels supplied in the challenge reference file.

Table 1. Season-level descriptive summary from the supplied surveillance data.

Season	States	Total cases	Total hosp.	Total deaths	Mean cases/100k
2023/2024 (Moderate)	37	17,115,135	267,587	24,443	144.4
2024/2025 (High)	37	31,416,391	455,557	42,412	263.2
2025/2026 (Moderate-High)	37	22,239,254	327,633	30,603	184.8

Regional structure in the most recent season (2025/2026)

Table 2. Zonal burden and mean healthcare-access index, 2025/2026 season.

Zone	Total cases	Mean incidence/100k	Mean healthcare access
NC	3,457,472	182.0	0.586
NE	2,930,880	172.4	0.395
NW	5,193,100	168.7	0.443
SE	2,944,649	206.5	0.702
SS	2,996,013	183.1	0.672
SW	4,717,140	202.9	0.733

Across all state-weeks in the data, the healthcare-access score exhibits almost no simple association with the reported case rate ($r = 0.013$) and only a weak negative association with the death rate ($r = -0.080$), whereas hospitalisation and death rates are, as expected, strongly associated with the case rate ($r = 0.91$ and $r = 0.87$, respectively). These are broad, state-level associations and should not be interpreted as evidence of individual-level or causal effects.

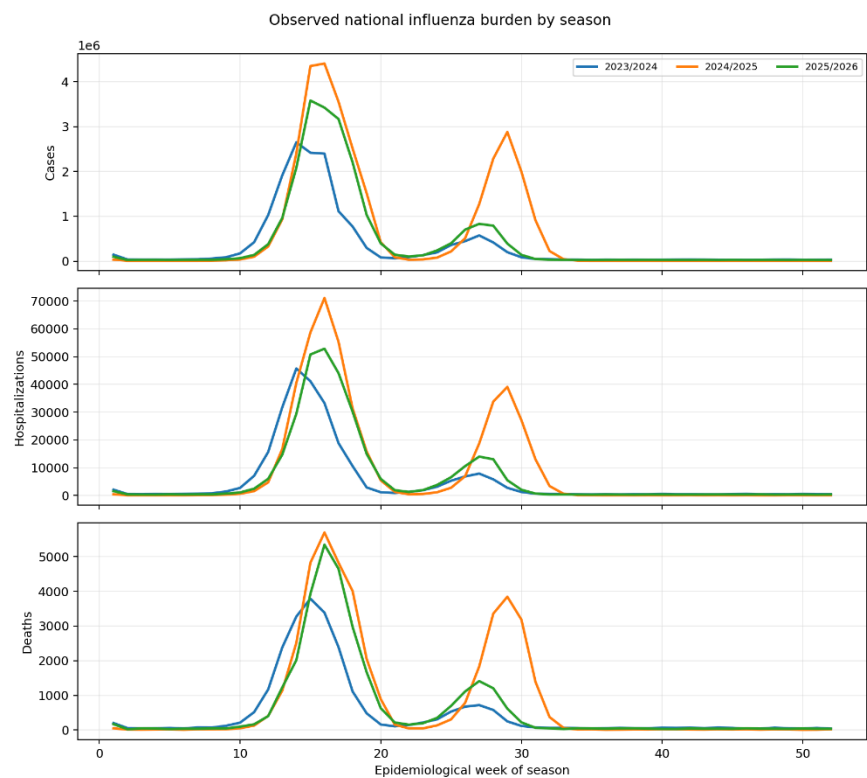


Figure 1. Observed national weekly cases, hospitalisations and deaths, all three seasons.

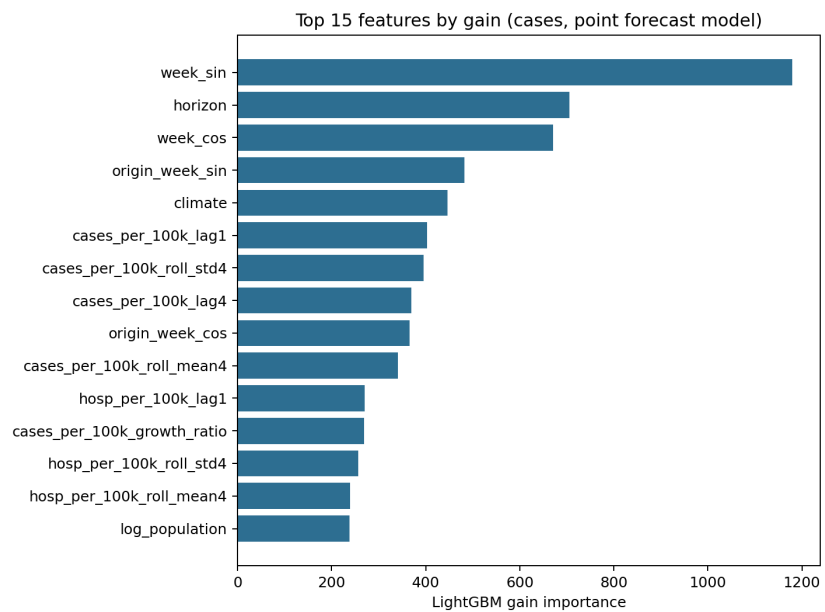


Figure 2. Top 15 features by gain, cases model (LightGBM component of the ensemble).

6.2 Feature importance

Ranking the LightGBM component's features by their contribution to error reduction (gain; Figure 2) indicates substantial reliance on the target week's calendar position (the sine-cosine pair), the forecast horizon, the forecast origin's calendar position, the state's climate class, and the most recent week of case history. This is a reassuring and interpretable result: the model relies principally on seasonal timing and recent momentum, the two signals a public-health analyst would consider first.

6.3 Rolling-origin evaluation

Each model was refitted using only the seasons preceding each test season, then evaluated against the two most recent seasons (2024/2025 and 2025/2026), forecasting eight weeks ahead from nine distinct origins per season. Table 3 reports the deployed ensemble alongside LightGBM alone, averaged across both test seasons.

Table 3. Rolling-origin evaluation, mean across both test seasons and all horizon weeks (state level; per 100,000 population).

Model	Outcome	MAE	RMSE	WIS*	CRPS	90% cov.
Ensemble	Cases	149.0	369.7	491.3	104.2	0.504
LightGBM only	Cases	148.6	358.7	542.7	113.1	0.437
Ensemble	Hospitalisations	2.17	5.08	6.84	1.46	0.502
LightGBM only	Hospitalisations	2.18	5.05	7.70	1.61	0.458
Ensemble	Deaths	0.216	0.501	0.652	0.141	0.621
LightGBM only	Deaths	0.215	0.501	0.713	0.152	0.668

Averaged over both test seasons, the ensemble's point accuracy is essentially equivalent to that of LightGBM alone (MAE within 0.5% for every outcome), while its probabilistic scores are consistently superior: WIS* decreases by 9.5% for cases, 11.1% for hospitalisations and 8.6% for deaths, with CRPS decreasing by a comparable margin.

These averages conceal a substantial and informative difference between the two test seasons. When trained on only one season of history, the ensemble's 90% interval for cases captured the observed value in only 14.0% of cases; with two seasons of training history, this rose to 86.7%, with comparable improvements for hospitalisations (16.8% to 83.6%) and deaths (42.0% to 82.2%). Point accuracy improved correspondingly, with cases MAE falling from 217.2 to 80.8 following the addition of a second training season.

6.4 Model comparison experiment

Five candidate models were trained and evaluated under identical conditions, using the same feature table, the same rolling-origin fold (trained on 2023/2024 and 2024/2025, tested on 2025/2026, the largest available fold), and the same scoring rules: a seasonal-naïve baseline (each state's most recent observed value carried forward), a Random Forest, scikit-learn's Gradient Boosting implementation, LightGBM, and the proposed equal-weight LightGBM-Random Forest ensemble.

Table 4. Five-model comparison on the largest rolling-origin fold (train: 2023/2024-2024/2025; test: 2025/2026).

Model	Outcome	MAE	WIS*	Cov. 90%	Train time
Seasonal-naive	Cases	274.4	1,875.5	0.947	instant
Random Forest	Cases	85.8	185.6	0.875	7 s
Gradient Boosting (sklearn)	Cases	83.8	187.0	0.923	119 s
LightGBM	Cases	82.6	186.0	0.811	11 s
Ensemble (LightGBM + RF)	Cases	81.0	174.6	0.872	17 s
Seasonal-naive	Hospitalisations	4.09	13.76	0.927	instant
Random Forest	Hospitalisations	1.29	2.67	0.839	7 s
Gradient Boosting (sklearn)	Hospitalisations	1.28	3.02	0.903	119 s
LightGBM	Hospitalisations	1.28	2.69	0.815	11 s
Ensemble (LightGBM + RF)	Hospitalisations	1.25	2.55	0.842	18 s
Seasonal-naive	Deaths	0.398	1.198	0.911	instant
Random Forest	Deaths	0.140	0.289	0.840	7 s
Gradient Boosting (sklearn)	Deaths	0.154	0.335	0.969	96 s
LightGBM	Deaths	0.137	0.283	0.851	10 s
Ensemble (LightGBM + RF)	Deaths	0.135	0.277	0.824	17 s

All four machine-learning candidates reduced MAE to approximately one-third of the naive baseline's across every outcome. Among them, the ensemble achieved the lowest MAE, WIS* and CRPS (48.7 for cases, 0.721 for hospitalisations and 0.077 for deaths) for all three outcomes. It did not outperform every alternative on every criterion: LightGBM alone achieved a lower RMSE for cases, and scikit-learn's Gradient Boosting achieved the best-calibrated 90% coverage for cases, hospitalisations and deaths, albeit at approximately ten times the training time and with a less favourable WIS*.

6.5 Eight-week forecast beyond the most recent observed week

The final ensemble, trained on all three available seasons, was used to forecast eight weeks beyond the most recently observed week, commencing at week 52 of the 2025/2026 season. The national point forecast begins at approximately 32,700 cases in the first forecast week, reflecting the still-elevated conclusion of the observed season, before declining sharply to approximately 10,300-10,700 cases per week for weeks 54 through 58, then rising to approximately 11,500 and 13,600 cases in weeks 59 and 60 as the earliest signal of the subsequent season's onset emerges. Hospitalisations and deaths follow the same trajectory (Table 5).

Table 5. National 8-week forecast from origin week 52 of the 2025/2026 season.

Epi. week	Point forecast	50% interval	90% interval	Hosp. (mean)	Deaths (mean)
53	32,676	24,795 - 43,470	14,685 - 62,463	580	68
54	10,509	7,905 - 13,856	5,943 - 25,906	211	26
55	10,259	7,851 - 13,388	5,896 - 24,044	205	26
56	10,404	7,978 - 13,519	5,985 - 23,418	204	26
57	10,419	8,076 - 13,433	6,156 - 22,745	203	26
58	10,718	8,416 - 13,672	6,304 - 23,130	205	23
59	11,477	8,799 - 14,660	6,689 - 27,460	238	26
60	13,612	10,232 - 17,445	7,364 - 37,217	277	37

The forecast trajectory captures the transition from an elevated end-of-season week into the subsequent season's quiescent trough, consistent with the pattern evident in Figure 1 across all three historical seasons. The full forecast dataset (all states, zones and the national total, every horizon week) accompanies this report as a CSV file.

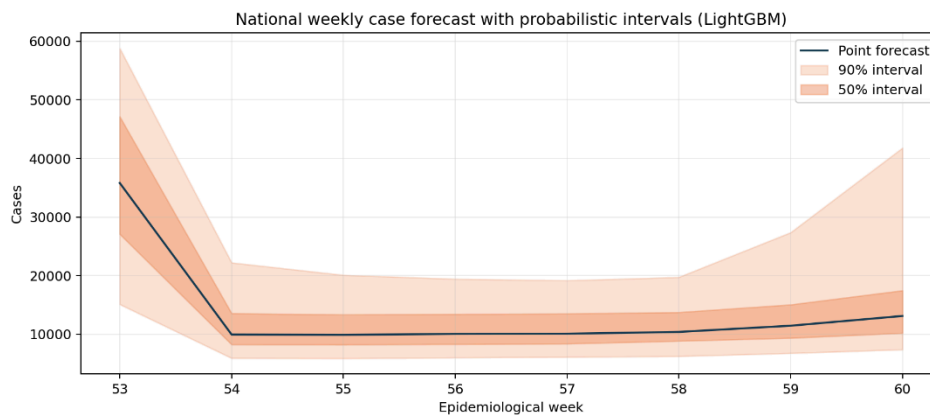


Figure 3. National weekly case forecast with 50% and 90% probabilistic intervals.

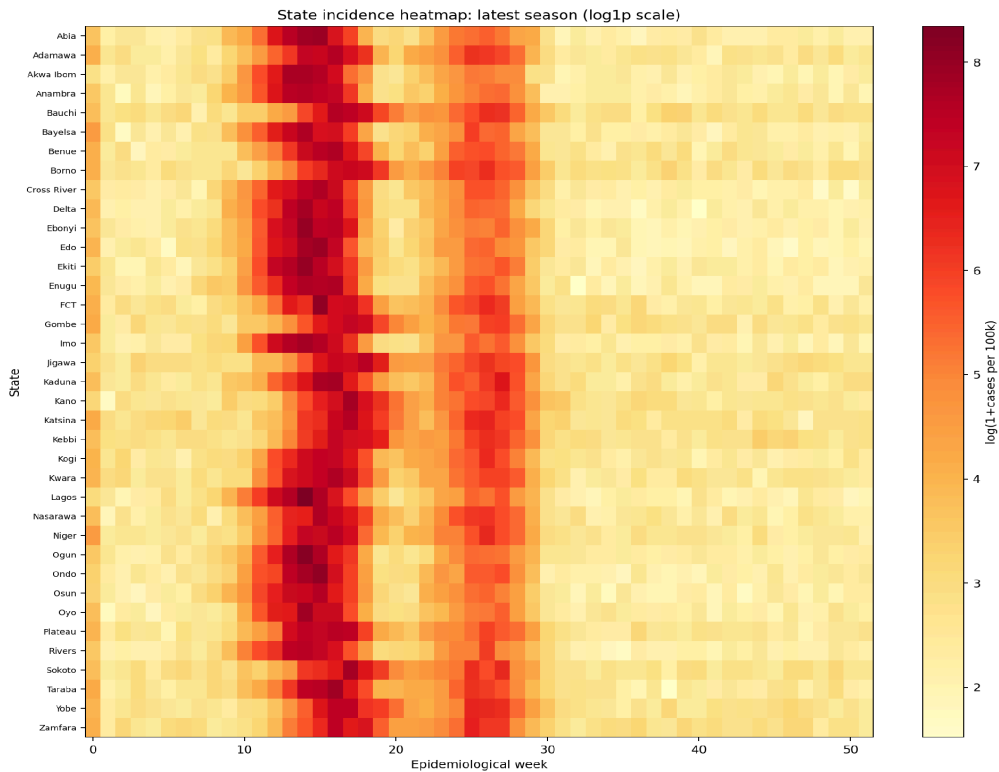


Figure 4. State-level incidence heatmap, most recent season (log1p scale).

6.6 Peak-week and transmission-direction probabilities

A Monte Carlo simulation derived from the ensemble's quantile forecasts assigns a 77.0% probability to the eight-week window's peak occurring in the first forecast week, consistent with the anticipated sharp decline from the still-elevated final observed week into the new season's trough; the remaining weeks each carry a comparatively small probability (2.0% to 7.1%). The probability that cases are increasing rather than decreasing from one week to the next begins very low (4.8% at the first step), remains close to an even 49-55% through weeks 55 to 59, and reaches 60.7% by week 60 as the earliest signal of the subsequent wave emerges.

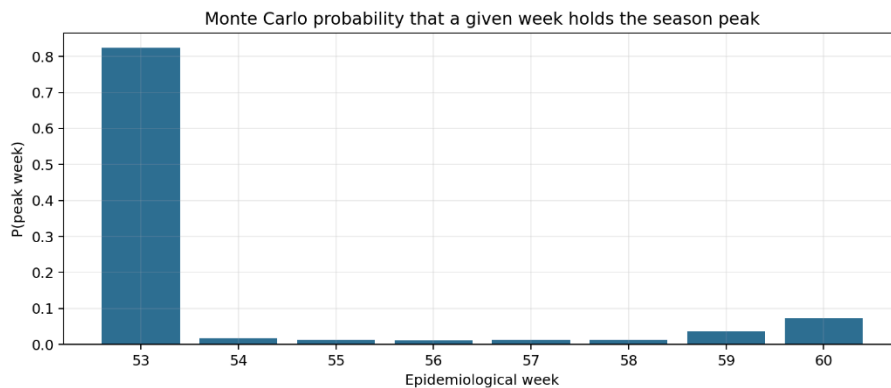


Figure 5. Monte Carlo probability that each horizon week holds the season peak.

7. Policy Implications

Forecasts accompanied by explicit 50% and 90% ranges, reported at state and zonal resolution, can support two practical planning decisions in advance of any observable rise in reported case numbers: the early positioning of surge capacity, staffing, bed capacity, antiviral supply and oxygen provision, in states whose forecast ranges indicate meaningful upside risk, and the targeted allocation of surveillance and outreach resources toward zones combining lower healthcare-access scores with substantial forecast burden. In the most recent season, the North-East and North-West zones exhibit the lowest mean healthcare-access scores in the dataset (0.395 and 0.443, respectively) alongside substantial case totals (2.93 million and 5.19 million cases), identifying them as reasonable priorities for strengthened reporting and response capacity. This conclusion holds notwithstanding the near-absence of a simple correlation between healthcare access and raw case numbers overall; lower access more plausibly affects the completeness of case ascertainment and the severity of outcomes than the underlying transmission rate.

A specific and material caveat for governance applications, revealed only through the rigorous evaluation undertaken in this study, is that the 90% intervals remain under-calibrated on average (50.4% measured coverage for cases across the two test seasons), notwithstanding an improvement to 86.7% once two seasons of training history became available. Pending implementation of the calibration remedy proposed in the Uncertainty Analysis section, planners are advised to interpret the reported ranges, particularly the comparatively wide national-level range, as indicative of relative risk and directional trend rather than as a guaranteed nine-in-ten bound, and to weight the point forecast and the overall trajectory of the forecast, whether rising, falling or stable, more heavily than the precise width of the associated interval.

The probabilistic peak-week and rising/falling-transmission outputs give planners a direct, quantified answer to two commonly asked operational questions, whether the peak is imminent and whether a week's rise is real or noise, without requiring informal interpretation of a single point forecast. These forecasts are intended to support, rather than substitute for, clinical and public-health judgement. The model does not establish a causal relationship between healthcare access and transmission, nor does it quantify the impact of any specific intervention; all forecasts represent patterns learned from three seasons of surveillance data.

8. Discussion

Interpretation of findings

The principal finding of this study is that a machine-learning approach is clearly warranted for this forecasting task: every tree-based model reduced MAE to approximately one-third of the naive baseline's, across all three outcomes, under a strictly out-of-sample evaluation. The second principal finding is that the ensemble constitutes the strongest all-round model. On the largest test fold it achieved the lowest MAE, WIS* and CRPS for every outcome, and across both test seasons it reduced WIS* by 9-11% relative to LightGBM alone while leaving MAE essentially unchanged. This pattern is consistent with expectations from the forecast-combination literature: averaging models whose errors are only partially correlated attenuates the component-specific errors, which improves probabilistic scores to a greater degree than the central forecast (Clemen, 1989; Timmermann, 2006; Wang et al., 2023).

The coverage results warrant particular attention. The ensemble's 90% interval coverage for cases rose from 14.0% to 86.7% solely through the addition of one further season of training history, with no change to the model, features or hyperparameters. This effect exceeds, by a considerable margin, the difference observed between any two of the five models compared in Table 4, suggesting that three seasons approximates the minimum training history required for a model of this kind to calibrate its uncertainty adequately, and that the calibration deficit reported here is substantially attributable to limited training data rather than to a fundamental deficiency in the modelling approach.

Strengths

1. The choice of model is supported by a controlled, same-data comparison rather than an assumption: four alternative models were actually trained and scored, and the result, a modest but consistent ensemble advantage rather than a decisive win, is reported without exaggeration.
2. Every machine-learning model evaluated comfortably and consistently outperformed a credible naive baseline across all outcomes.
3. The deployed ensemble is computationally inexpensive: it retrains in under twenty seconds per outcome, requires no tuned ensemble weights, and is served through a single shared feature-engineering module, one compressed model file, and a lightweight deployment service.
4. Evaluation is conducted strictly out-of-sample and in chronological order throughout, such that no test-season information is permitted to leak into a model trained to forecast that season.

Limitations

5. Interval coverage is the model's most significant limitation: 50.4% for cases against a 90% nominal target, averaged across the two test seasons.
6. The five-model comparison employs a single rolling-origin fold, the largest and most recent available, in order to maintain tractable computation time.
7. Random Forest and scikit-learn's Gradient Boosting were supplied with one-hot-encoded categorical features rather than LightGBM's native categorical handling, the conventional approach for those libraries but not a strictly equivalent comparison.
8. The ensemble weights are fixed at equal values and were not tuned.
9. National- and zone-level forecast ranges are constructed by summing state-level ranges, which assumes that states move in close to lockstep and consequently overstates the combined uncertainty.
10. Only three seasons of surveillance data are currently available, such that both the model's hyperparameters and the conclusions of the comparison experiment rest on a limited number of independent annual observations.

Why the ensemble, despite the gaps

The ensemble was selected for deployment because it represents the strongest all-round performer on the criteria of greatest relevance to operational forecasting, namely accuracy and probabilistic quality, at negligible additional computational cost. This is a more modest claim than an assertion that the ensemble is universally superior: it does not achieve the lowest RMSE for cases or the best-calibrated coverage, and the most consequential remaining limitation, interval calibration, is not resolved by ensemble alone.

9. Uncertainty Analysis

Four distinct factors jointly determine the width and calibration quality of the reported prediction ranges:

1. Calibration error within the quantile forecasts: each LightGBM quantile level is fitted as an independent model, such that nothing constrains the resulting quantiles to jointly describe one coherent distribution, and the Random Forest quantiles are themselves an approximation derived from tree-level disagreement rather than a complete quantile regression forest. This is likely the largest single contributor to the observed coverage deficit.
2. Small-sample instability: with only three seasons available, one rolling-origin fold is trained on a single season and the other on two; the substantial variation in coverage between these folds (for example, 9.3% to 78.1% for LightGBM's own cases coverage) demonstrates that the uncertainty estimate is highly sensitive to the quantity of historical data observed.
3. Fixed, untuned hyperparameters and ensemble weights: tree depth, learning rate, forest size and the ensemble weighting were each set once, at conventional values, in order to avoid overfitting a limited dataset. The reported results should not be interpreted as the ceiling of performance achievable by these model families.
4. Aggregation by summation: national- and zone-level ranges, constructed by summing state-level ranges, inherit and likely amplify the state-level calibration deficit rather than correcting it.

Collectively, these considerations point to a single, high-priority remedy: conformal calibration (Romano, Patterson, & Candes, 2019), a procedure that recalibrates a model's existing quantile predictions using a held-out calibration set such that the stated coverage is empirically achieved, without requiring additional training data or a different model family. A secondary, complementary remedy is to replace the summation-based aggregation of state-level ranges with a joint Monte Carlo simulation that preserves correlation structure across states, which would narrow the national- and zone-level ranges without altering the underlying state-level forecasts.

10. Conclusions

This study developed a machine-learning forecasting system for Nigeria's influenza surveillance data and evaluated its design empirically rather than adopting assumptions drawn from external literature. A seasonal-naive baseline, a Random Forest, scikit-learn's Gradient Boosting implementation, LightGBM, and an equal-weight LightGBM-Random Forest ensemble were trained and scored on identical data under an identical protocol. All machine-learning candidates substantially outperformed the naive baseline; the ensemble achieved the best MAE, WIS* and CRPS for cases, hospitalisations and deaths on the largest test fold, and reduced WIS* by 9-11% relative to LightGBM alone across both test seasons, although it did not outperform every alternative on every metric, and its interval coverage, while improving markedly with additional training data, remains below its nominal target.

11. Recommendation

Four recommendations follow directly from this study's findings. First, apply conformal calibration to the ensemble's quantile forecasts before relying on stated coverage for operational planning. Second, retrain the model as each new season becomes available, given that the increase in cases coverage from 14.0% to 86.7% after one additional season is the strongest evidence in this study that more data, not a different model, offers the most direct route to improved calibration. Third, extend the five-model comparison across every rolling-origin fold once sufficient seasons have accumulated, and evaluate

whether a trained or stacked ensemble outperforms the present equal-weight average, potentially including scikit-learn's Gradient Boosting as a third member given its superior calibrated coverage. Fourth, replace the summation-based aggregation of state-level ranges with a joint Monte Carlo simulation that preserves cross-state correlation structure.

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